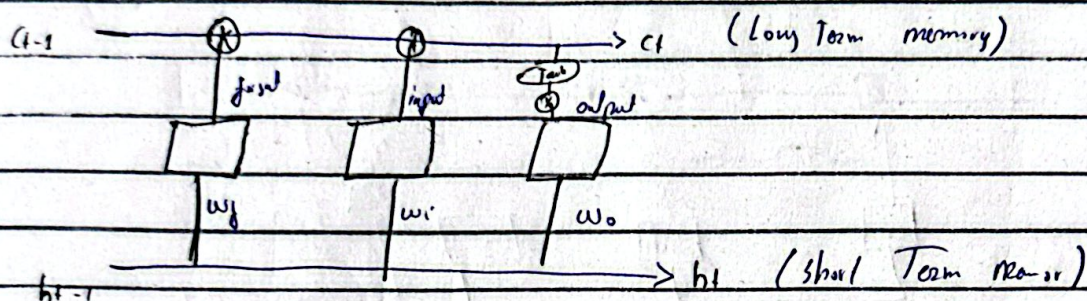


⇒ GRU RNN

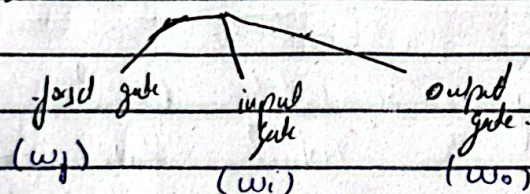
★ Gated Recurrent unit RNN.

★ Developed in 2014

★ Ordinary LSTM RNN.

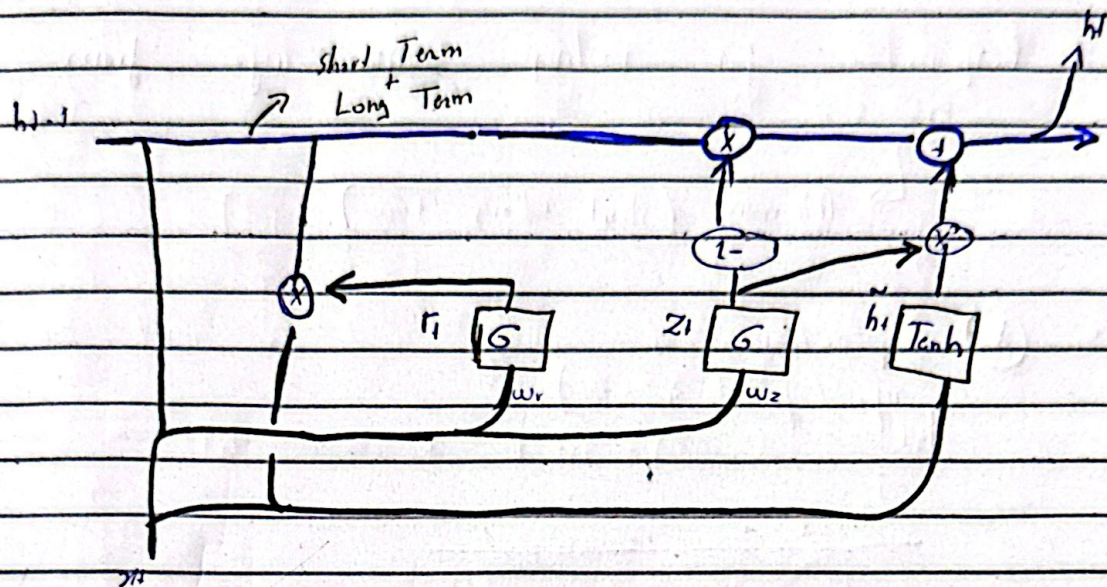


⇒ The Architecture is Complex.



⇒ if each gate has 100s of neurons
so there are weights & biases for each. → Means there are more trainable parameters → means more time.

*** In GRU both Long Term & Short Term memory is Combined.



↳ Update gate.

⇒ 1st Find $z_t \Rightarrow$ Update gate

$$z_t = \sigma(w_z \cdot [h_{t-1}, x_t])$$

↳ Reset Gate.

⇒ Then find $r_t \Rightarrow$ Reset Gate.

$$r_t = \sigma(w_r \cdot [h_{t-1}, x_t])$$

↳ Temporary Hidden State:

⇒ Find $\tilde{h}_t$.

$$\tilde{h}_t = \tanh(W \cdot [r_t \otimes h_{t-1}, x_t])$$

↳ Final state:

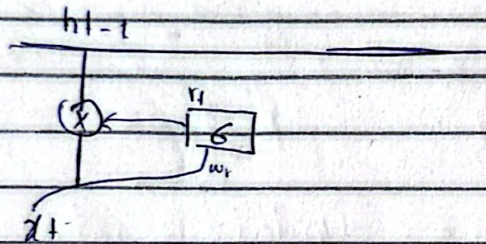
$$h_t = (1 - z_t) \otimes h_{t-1} + z_t \otimes \tilde{h}_t$$

⇒ Reset Gate:

⇒ Responsible for resetting some info from h_{t-1} .

→ Memory (Short + Long Term).

- $(\times)$ operation is always used to reset some info. (forget or add).



$$\Rightarrow r_t = \sigma(w_r [h_{t-1}, x_t])$$

e.g. $h_{t-1} = [0.6 \quad 0.5 \quad 0.3 \quad 0.4]$
 $r_t = [0.2 \quad 0.4 \quad 0.8 \quad 0.2]$

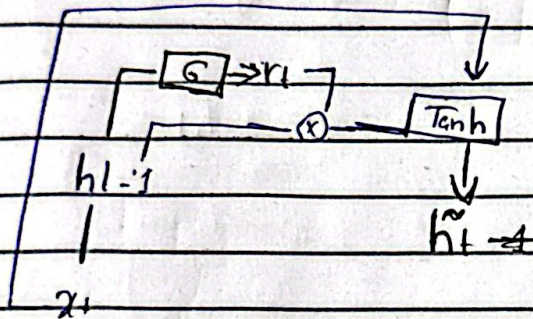


$$x_t \rightarrow [0.12 \quad 0.20 \quad 0.24 \quad 0.18]$$

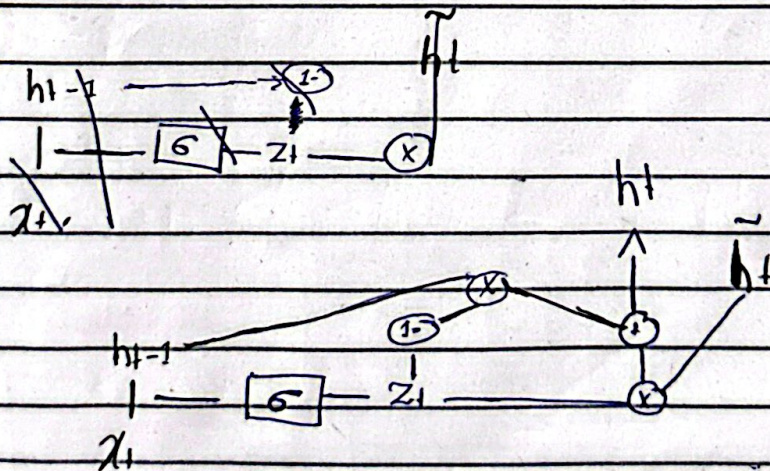
- So I have reset the value based on content.

A Then we pass the reset value to Tanh.

⇒ Update gate.



Now To go from $\tilde{h}_t$ to h_t is most imp step



* Observe we have coupled Update gate &
Temporary hidden state.

*** Update gate: What context info should
be added.

→ but This is completely dependent on
Temporary hidden state / Candidate state.

→ Candidate gate has current context. ($\tilde{h}_t$)

* if current context is imp ($\tilde{h}_t$), then I
will add less info from (z_t).

$$h_t = \underbrace{(1 - z_t) * h_{t-1}}_{\text{New Context}} + \underbrace{z_t * \tilde{h}_t}_{\text{Current Context}}$$